SHAULI RAVFOGEL

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Faculty Fellow, New York University Center of Data Science. My research interests encompass representation learning, unsupervised machine learning, causal attribution, and interpretability of neural models for NLP.

EDUCATION

- **Sep 2020-Sep 2024** PhD student in Computer Science, Bar-Ilan University.
  Dissertation title: Analyzing the representation space of transformer-based language models.
  Supervisor: Prof. Yoav Goldberg.
- **Sep 2018-Sep 2020:** MSc in Computer Science, Bar-Ilan University, supervised by Prof. Yoav Goldberg.
- **Oct 2015-Sep 2018:** BSc in Computer Science, Bar-Ilan University.
- **Oct 2010-Sep 2013:** BSc in Chemistry, Bar-Ilan University.

AWARDS

- IAAI Best PhD Thesis Award (2024).
- Israeli Council for Higher Education (Vatat) Postdoctoral Fellowship in Data Science (2024).
- Fulbright Postdoctoral Fellowship (2024; declined after accepting an offer from the NYU CDS Faculty Fellowship).
- Blavatnik Prize for Outstanding Israeli Doctoral Students in Computer Science (2024).
- DAAD AInet Postdoctoral Fellowship (2023).
- Bloomberg Data Science PhD Fellowship (2022–2024; first recipient outside of the US).
- Israeli Council for Higher Education (Vatat) Fellowship for outstanding PhD Students in Data Sciences.
- Outstanding Paper Award, Bar-Ilan Data Science Institute (2023).
- Bar-Ilan Presidential Scholarship for Outstanding PhD Students (2021–2024).
- Best Paper Award, RepL4NLP workshop (2022).

GRANTS AND FUNDING

- **Co-Principal Investigator**, *LLM Introspection.* Arena Academic Partnerships Program (with Tal Linzen). NYU Center for Data Science, 2026.

PROFESSIONAL EXPERIENCE AND APPOINTMENTS

- **Jan 2025-** Faculty Fellow, NYU
- **September-December 2024**: Research visit, ETH Zurich.
- **October 2023-September 2024** Student Researcher, Google Research
- **June-September 2023:** Research internship, Bloomberg London.
- **June-September 2022:** Research internship, Facebook AI Research, Israel.

- **January 2020-summer 2022 & Autumn 2022-summer 2023:** Ph.D. Research Intern, AI2 Israel.

TEACHING EXPERIENCE

- **Lecturer:** Causality and Interpretability of Language Models, NYU (Winter 2025)
- **Lecturer:** Introduction to Machine Learning, Bar-Ilan University (2020–2021)
- **Teaching Assistant:** Graduate Machine Learning Course, Bar-Ilan University (2019–2021)

LIST OF PUBLICATIONS

* denotes equal contribution.

Under submission

- John Morrison, Mouad Oulouali, Eivinas Butkus, Samuel Lippl, **Shauli Ravfogel**. "What Counts as Evidence for a World Model? Reassessing Othello-GPT."
- Zhuofan Josh Ying, **Shauli Ravfogel**, Nikolaus Kriegeskorte, Peter Hase. "The Truthfulness Spectrum Hypothesis."
- Hung-Ting Chen, Xiang Liu, **Shauli Ravfogel**, Eunsol Choi. "Beyond Single Embeddings: Capturing Diverse Targets with Multi-Query Retrieval."
- Royi Rassin, Aviv Slobodkin, **Shauli Ravfogel**, Yanai Elazar, Yoav Goldberg. "GRADE: Quantifying Sample Diversity in Text-to-Image Models."

Accepted

First/Last Author:

- **Shauli Ravfogel**, Gilad Yehudai, Joan Bruna and Alberto Bietti. "Geometric Factual Recall in Transformers." Advances in Neural Information Processing Systems (NeurIPS) 2026.
- Shashwat Singh, Tal Linzen, **Shauli Ravfogel**. "Can LLMs Introspect? A Reality Check." Conference on Language Modeling (COLM) 2026.
- **Shauli Ravfogel**, Gilad Yehudai, Tal Linzen, Joan Bruna and Alberto Bietti. "Emergence of Linear Truth Encodings in Language Models." Advances in Neural Information Processing Systems (NeurIPS) 2025.
- Floris Holstege*, **Shauli Ravfogel***, Bram Wouters. "Preserving Task-Relevant Information Under Linear Concept Removal." Advances in Neural Information Processing Systems (NeurIPS) 2025.
- **Shauli Ravfogel**, Anej Svete*, Vésteinn Snæbjarnarson, Ryan Cotterell. "Gumbel Counterfactual Generation from Language Models." ICLR 2025.
- Matan Avitan, Ryan Cotterell, Yoav Goldberg, and **Shauli Ravfogel**. "Natural Language Counterfactuals through Representation Surgery." Findings of NAACL 2025.
- **Shauli Ravfogel**, Valentina Pyatkin, Amir David Nissan Cohen, Avshalom Manevich, Yoav Goldberg. "Description-based Text Similarity." COLM 2024.
- Shashwat Singh*, **Shauli Ravfogel***, Roee Aharoni, Jonathan Herzig, and Ponnurangam Kumaraguru. "Representation Surgery: Theory and Practice of Affine Steering." ICML 2024.

- Aviv Slobodkin, Omer Goldman, Avi Caciularu, Ido Dagan, **Shauli Ravfogel**. "The Curious Case of Hallucinatory Unanswerability: Finding Truths in the Hidden States of Over-Confident Large Language Models." EMNLP 2023.
- **Shauli Ravfogel**, Yoav Goldberg, and Ryan Cotterell. "Log-linear Guardedness and its Implications." In Proceedings of ACL 2023.
- **Shauli Ravfogel**, Yoav Goldberg, and Jacob Goldberger. "Conformal Nucleus Sampling." Findings of ACL 2023.
- **Shauli Ravfogel**, Michael Twiton, Yoav Goldberg, and Ryan D. Cotterell. "Linear Adversarial Concept Erasure." In International Conference on Machine Learning (ICML) 2022.
- **Shauli Ravfogel**, Francisco Vargas, Yoav Goldberg and Ryan Cotterell. "Kernelized Concept Erasure." In Proceedings of EMNLP 2022.
- Royi Rassin, **Shauli Ravfogel***, and Yoav Goldberg. "DALLE-2 is Seeing Double: Flaws in Word-to-Concept Mapping in Text2Image Models." In Proceedings of BlackboxNLP 2022.
- **Shauli Ravfogel**, Grusha Prasad, Tal Linzen, and Yoav Goldberg. "Counterfactual Interventions Reveal the Causal Effect of Relative Clause Representations on Agreement Prediction." In Proceedings of CoNLL 2021.
- Carlo Meloni*, **Shauli Ravfogel***, and Yoav Goldberg. "Ab Antiquo: Neural Proto-language Reconstruction." In Proceedings of NAACL 2021.
- **Shauli Ravfogel**, Yanai Elazar, Jacob Goldberger, and Yoav Goldberg. "Unsupervised Distillation of Syntactic Information from Contextualized Word Representations." In Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP 2020.
- **Shauli Ravfogel**, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. "Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection." In Proceedings of ACL 2020.
- **Shauli Ravfogel**, Yoav Goldberg, and Tal Linzen. "Studying the Inductive Biases of RNNs with Synthetic Variations of Natural Languages." In Proceedings of NAACL 2019.
- **Shauli Ravfogel**, Yoav Goldberg, and Francis Tyers. "Can LSTM Learn to Capture Agreement? The Case of Basque." In Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP.

Co-author:

- Or Shafran, Shaked Ronen, Omri Fahn, **Shauli Ravfogel**, Atticus Geiger, and Mor Geva. "From Directions to Regions: Decomposing Activations in Language Models via Local Geometry." ICML 2026.
- Itamar Trainin, **Shauli Ravfogel**, Omri Abend, Amir Feder. "Discrete Diffusion Models Exploit Asymmetry to Solve Lookahead Planning Tasks." Advances in Neural Information Processing Systems (NeurIPS) 2026.
- Mosh Levy, Zohar Elyoseph, **Shauli Ravfogel**, and Yoav Goldberg. "State over Tokens: Characterizing the Role of Reasoning Tokens." Nature Machine Intelligence, 2026.
- Jackson Petty, Michael Y. Hu, Wentao Wang, **Shauli Ravfogel**, William Merrill, Tal Linzen. "RELIC: Evaluating Complex Reasoning via the Recognition of Languages In-Context." Transactions of the Association for Computational Linguistics (TACL) 2026.
- Aviya Maimon, Amir David Nissan Cohen, Gal Vishne, **Shauli Ravfogel**, Reut Tsarfaty. "From Benchmarks to Skills: Low-Rank Factors for LLM Evaluation." Transactions of the Association for Computational Linguistics (TACL) 2026.

- Yu Fan, Yang Tian, **Shauli Ravfogel**, Mrinmaya Sachan, Elliott Ash, and Alexander Miserlis Hoyle. "The Medium Is Not the Message: Deconfounding Document Embeddings via Linear Concept Erasure." In Proceedings of EMNLP 2025, pp. 32112–32131.
- Yihuai Hong, Lei Yu, **Shauli Ravfogel**, Haiqin Yang, Mor Geva. "Intrinsic Evaluation of Unlearning Using Parametric Knowledge Traces." In Proceedings of EMNLP 2025.
- Marius Mosbach, Tiago Pimentel, **Shauli Ravfogel**, Dietrich Klakow, and Yanai Elazar. "Few-shot Fine-tuning vs. In-context Learning: A Fair Comparison and Evaluation." Findings of ACL 2023.
- Royi Rassin, Eran Hirsch, Daniel Glickman, **Shauli Ravfogel**, Yoav Goldberg, and Gal Chechik. "Linguistic Binding in Diffusion Models: Enhancing Attribute Correspondence through Attention Map Alignment." NeurIPS 2023.
- Nora Belrose, David Schneider-Joseph, **Shauli Ravfogel**, Ryan Cotterell, Edward Raff, and Stella Biderman. "LEACE: Perfect Linear Concept Erasure in Closed Form." NeurIPS 2023.
- Mosh Levy, **Shauli Ravfogel**, Yoav Goldberg. "Guiding LLM to Fool Itself: Automatically Imbuing Multiple Shortcut Triggers in Question Answering Samples." Findings of EMNLP 2023.
- Rita Sevastjanova, Eren Cakmak, **Shauli Ravfogel**, Ryan Cotterell, and Mennatallah El-Assady. "Visual Comparison of Language Model Adaptation." IEEE Transactions on Visualization and Computer Graphics 2022.
- Hila Gonen, **Shauli Ravfogel**, and Yoav Goldberg. "Analyzing Gender Representation in Multilingual Models." Proceedings of the 7th Workshop on Representation Learning for NLP 2022 (Best Paper Award).
- Elad Ben-Zaken, **Shauli Ravfogel**, and Yoav Goldberg. "BitFit: Simple Parameter-Efficient Fine-Tuning for Transformer-Based Masked Language-Models." In Proceedings of ACL 2022.
- Yanai Elazar, Nora Kassner, **Shauli Ravfogel**, Abhilasha Ravichander, Eduard Hovy, Hinrich Schütze, and Yoav Goldberg. "Measuring and Improving Consistency in Pretrained Language Models." Transactions of the Association for Computational Linguistics 2021.
- Yanai Elazar, **Shauli Ravfogel**, Alon Jacovi, and Yoav Goldberg. "Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals." Transactions of the Association for Computational Linguistics 2021.
- Alon Jacovi, Swabha Swayamdipta, **Shauli Ravfogel**, Yanai Elazar, Yejin Choi, and Yoav Goldberg. "Contrastive Explanations for Model Interpretability." In Proceedings of EMNLP 2021.
- Hila Gonen, **Shauli Ravfogel**, Yanai Elazar, and Yoav Goldberg. "It's not Greek to mBERT: Inducing Word-Level Translations from Multilingual BERT." In Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP 2020.

ACADEMIC ACTIVITY

- **Service:** Senior Area Chair, ACL Rolling Review and AAAI 2027 AI Alignment Track (SPC); Area Chair, ICLR, EMNLP, COLM; Reviewer, ACL, EMNLP, NAACL, NeurIPS, ICML; Grant reviewer, U.S.–Israel Binational Science Foundation (BSF).
- **Organization & mentoring:** Co-organizer, RepL4NLP Workshop at ACL 2023 (250+ attendees); co-advised several master's students.
- **Selected invited talks:** UPenn NLP (2026); Aspen Meeting on Foundation Models, UK AISI Alignment Conference, Cambridge LTL (2025); Technion (2024); EPFL, DeepMind, J. P. Morgan, IBM Research, Hebrew University (2023); SIGTYP Lecture Series (2021); Yale, Technion (2020).

- **Research visits:** Prof. Ryan Cotterell's group, ETH Zürich (three stays, 2021–2023); Prof. Tal Linzen's group, Johns Hopkins University (2020).